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This time it's different – Generative artificial intelligence and occupational choice

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ABSTRACT

We show the causal influence of the launch of generative artificial intelligence (AI) in the form of ChatGPT on the search behavior of young people for apprenticeship vacancies. To estimate the short- and medium-term effects, we use a variety of methods, including a difference-in-discontinuity approach exploiting the exogenous nature of the unanticipated launch of ChatGPT in 2022. There is a strong short- and medium-term decline in the intensity of searches for vacancies, indicating a notable reduction in the supply of young people actively seeking apprenticeships and suggesting great uncertainty among the affected cohort. Occupations with a high proportion of cognitive tasks and with high demands on language skills were particularly affected by the decline. Interestingly, the revealed preferences in the search behavior of young job seekers contrasted with previous expert assessments on the automation risks of occupations and aligned with the most recent assessments of the AI and language model exposure of occupations – before these new assessments existed. Notably, while the supply decline did not reduce the number of signed apprenticeship contracts, we find evidence of declining applicant quality, particularly for commercial employees, the most widely offered apprenticeship in Switzerland.

1. Introduction

The announcement and public deployment of the generative artificial intelligence (AI) ChatGPT version 3.5 led to a shock shift in perceptions of how AI can impact future workplaces. In occupations that felt safe from being partially or entirely replaced, the risk of automation has suddenly increased considerably because, unlike in the past, when the risk of automation was equated with the replacement of manual and routine labor, the emerging generative AI is targeting non-routine and cognitive tasks.

In this study, we show how the unexpected launch of ChatGPT in November 2022 had an immediate, significant, and long-lasting causal effect on young people's search behavior for apprenticeship vacancies. The impact on search behavior for specific occupations is of great economic importance as it involves a high-stakes decision. Apprentices invest a large part of their three to four years of training in learning occupation-specific skills that enable them to perform specialized tasks. If technology can easily perform these tasks now or in the near future, this investment will be lost and must be replaced by further investment in human capital. Therefore, if the perceived impact of ChatGPT is sufficiently salient and credible, it is reasonable that potential apprentices are already considering the resulting changes in expected demand for specific occupations when searching and deciding on a training occupation. For example, a recent study suggests that young people in the US have increased their college enrollment in response to the introduction of industrial robots, a technology that requires significant investment and time to implement (Di Giacomo and Lerch, 2023). In contrast, generative AI is cheap, widely accessible, and rapidly expanding, making its potential impact broader and more immediate.

In the past, observations of skill shifts in industrialized labor markets showed, first and foremost, that digitalization was skill-biased (Katz and Murphy, 1992), shifting demand from less skilled to more skilled workers and leading to a skill upgrading of the labor force in developed economies (Berman et al., 1998). Second, digital technology increasingly substituted workers in routine-intensive jobs while complementing workers executing non-routine tasks (Autor et al., 2003). As a result, many labor markets experienced a decline in routine employment and steady growth in non-routine cognitive jobs, which predominantly require higher levels of education, further contributing to the ongoing upgrading trends in developed economies (e.g., Goos and Manning,

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2007; Goos et al., 2009; Gschwendt, 2022). As such, advances in digital technologies can be considered a key driver of widespread skill upgrading (e.g., Katz and Autor, 1999; Acemoglu and Autor, 2011; Oesch and Rodríguez Menés, 2011; Balsmeier and Woerter, 2019), and, in turn, were more beneficial for skilled than for unskilled workers, as occupations became more demanding and complex and thus requiring longer, higher, and better training.

Considering the labor market effects of generative AI, however, initial observations and estimates point in a different direction than previous trends with regard to specific skills and tasks. First empirical studies show that the use of (generative) AI leads to increased productivity in more complex and less ambiguous tasks, with a greater positive impact on low-skilled than on high-skilled workers (Acemoglu et al., 2022; Haslberger et al., 2023; Brynjolfsson et al., 2023). While current analyses of job ads do not yet show a high prevalence of jobs in which AI skills are used (Green, 2024), early estimates suggest that many occupations will be affected by AI in the near future, with higher-educated workers disproportionally exposed (Eloundou et al., 2023).

Early generative AI models, such as GPT-3.5, used in the initial release of ChatGPT, demonstrated notably stronger capabilities in tasks demanding literacy/reading compared to mathematical skills. This is relevant for potential labor market effects because while only between 27 % and 44 % of workers daily perform numeracy tasks at work, having numeracy proficiency below or at the level of AI, almost 60 % of the workforce using literacy skills daily have a proficiency comparable to or below that of computers (for numbers for countries having participated in the PIAAC survey, see OECD, 2023).

Since its release in 2022, ChatGPT has significantly improved in mathematical skills and across other fields of usage, such as reasoning, while several other advanced large language models (LM) have also been launched and continuously enhanced (Korinek, 2024a). While earlier AI systems were only able to perform individual or a small range of tasks, large LMs are increasingly general-purpose, able to perform a growing range of cognitive tasks increasingly well. Some economists consider large LMs as precursors to artificial general intelligence by some economists, which would create the potential for widespread substitution of human cognitive labor in the long term (Korinek, 2024b). However, as the first empirical studies show, ChatGPT and other generative AI models have already negatively affected labor demand in exposed occupations in the short term. Hui et al. (2024) find that the availability of ChatGPT, DALL-E-2, and Midjourney, two image-generating AI models, significantly reduced the employment and earnings of exposed freelancers, with a greater negative impact on freelancers who produced higher-quality work before the tools were introduced. Similarly, Demirci et al. (2025) show that postings for exposed jobs decreased after the introduction of ChatGPT and image-generating AI technologies, respectively. These findings provide initial evidence of the potentially disruptive impact of generative AI on labor markets.

Our study complements the existing literature in two ways. First, by investigating the supply-side effects of the new generative AI, i.e., how it affects the perceptions of tomorrow's workforce and their occupational choices. Second, by examining this for a high-stakes and long-term decision, i.e., to learn specialized occupation-specific skills for three or four years. We employ a variety of methods to investigate both the immediate, short-term, and medium-term effects. A regression discontinuity design allows us to treat the exogenous nature of the introduction of ChatGPT close to its launch as a quasi-experiment. We conduct a difference-in-discontinuity approach to exclude any seasonal influences in the labor supply that bias the results, taking previous, but unaffected, labor supply underlying the same seasonal patterns as a counterfactual. Complementary, a month fixed-effects estimation helps us to investigate the medium-term impacts of generative AI on labor supply.

We find that with the launch of ChatGPT, the supply of potential apprentices, in the form of search queries for apprenticeship positions, decreases substantially by, on average, about 8 percent, in the short- and medium-term. While this speaks to a general uncertainty about which skills will be made redundant by generative AI in the future, we see substantial heterogeneity for different occupations according to cognitive requirements and task types performed in these occupations. Occupations involving primarily cognitive tasks are particularly affected by the shock in perception, with the largest decline in supply. Similarly, those occupations with the highest language requirements experienced the sharpest decrease in supply – an area of expertise in which, at first glance, a large language model such as ChatGPT is most likely to replace jobs. In a descriptive analysis on the tangible effects of the decrease in labor supply on the equilibrium, we find a negative, but insignificant correlation of the launch of ChatGPT on the number of signed contracts and, in a case study on the commercial employee, a decrease in the average ability in the applicant pool after ChatGPTs launch. All of this contrasts with previous waves of automation, which we can also see from the fact that for the occupations with the highest automation risk attributed in earlier expert assessments, we see the smallest reduction in supply: This time, it's different.

The remainder of the paper is structured as follows: In the next section, we discuss the interplay between the launch of ChatGPT and how the apprenticeship market works. In Section 3, we describe the data used, and in Section 4, the empirical strategy. Section 5 shows the empirical results, and Section 6 concludes with a discussion.

2. The launch of ChatGPT and the Swiss apprenticeship market

With the announcement and public release of version 3.5 of ChatGPT[1] on November 30th, 2022, the public at large has been granted access to highly capable generative artificial intelligence for the first time. Unlike previous disruptive technologies, it was instantly accessible for anyone with internet access to test and use. The opportunity met with overwhelming interest: ChatGPT reached one million users within 5 days. Intense media coverage and debate on social media quickly followed, leading to a major shift in public perception of the possibilities of AI almost overnight.

This increased attention to the possibilities of AI hit Swiss adolescents in the midst of making a far-reaching, high-stakes decision about their educational and occupational future. After compulsory school, approximately 70 % of young people in Switzerland start a three- or four-year apprenticeship and choose among about 200 different occupations. In an apprenticeship, apprentices acquire general, but most importantly occupation-specific skills, spending most of their time in their company and 1–2 days a week in a vocational school. While some of these occupation-specific skills can be transferred to other occupations (Eggenberger et al., 2022), there is a considerable risk that the significant time and financial investment of an apprenticeship may become partially worthless if some of the skills learned can be easily substituted by AI now or in the near future.

The Swiss apprenticeship market operates in many ways like a regular labor market. Companies create demand by posting apprenticeship vacancies, typically starting in August of the year before apprenticeships begin, which is usually between July and September.[2] Young people interested in apprenticeships begin searching and applying about a year in advance, with first contracts often being signed as early as October or November, and activity peaking in the spring before the new school year. In other words, the launch of ChatGPT coincided with the time when most young people interested in an apprenticeship were very concretely considering their choice of a specific occupation.

The process of finding an apprenticeship is competitive and mirrors broader job market dynamics. Applicants submit, on average, just over

[1] When referring to "ChatGPT", we always mean ChatGPT version 3.5.

[2] The interested reader is referred to, e.g., Wolter and Ryan (2011) or Muehlemann and Wolter (2020), for an overview of the economics of the apprenticeship market.

eight applications before securing an apprenticeship (GFS Bern, 2024). While the overall number of apprenticeship positions in recent years has been sufficient for all young people to secure a place, not everyone is able to obtain their preferred position. There are both attractive and less attractive occupations and employers, leading to varying levels of competition: for highly sought-after apprenticeships, the ratio of successful applicants to total applicants can reach 1:20 or more, whereas for less popular positions, a single application may suffice. The timing and speed at which positions are filled depend on regional differences, the popularity of the occupation, and the level of demand for apprentices. Once both parties agree, a work and training contract is signed, and all apprenticeships begin after the summer school holidays.

In summary, the Swiss apprenticeship market is seasonally driven, highly structured, and functions with competitive dynamics similar to the broader labor market, all of which shaped the context in which Swiss adolescents encountered the disruptive potential of AI technologies like ChatGPT.

3. Data

For our research question, precisely tracking young people's career choice behavior over time is crucial. This is possible because we can use data from the national platform for apprenticeship vacancies. The majority of training companies operating in Switzerland advertise their apprenticeship vacancies on this online platform.[3] Interested young people can use this platform to search for vacancies in specific occupations within a certain geographical perimeter and thus obtain information on which apprenticeships are available at which companies.[4] If they are interested, they can also find the training companies' contact details so they can apply. Therefore, the search behavior on this platform is not a mere search for information about occupations (for which other platforms exist); it shows a concrete interest in finding an apprenticeship in a specific occupation. Moreover, and this is crucial for our empirical strategy, the information about which company currently has vacancies in the occupation one is interested in cannot be obtained by asking ChatGPT because the generative AI did not have access to real-time information, at least not at the time of the study. In sum, the number of search queries on this platform provides a timely and occupation-specific proxy for the supply side of the apprenticeship market, capturing the pool of young people actively seeking apprenticeship positions. All search queries on this platform are recorded and used in this paper in an aggregated form on a daily basis, showing the number of queries by region and occupation.[5]

For this research, we extracted the number of search queries from log files from the 1st of January 2021 until the 30th of June 2023, resulting in 4,541,290 observations. The outcome variable represents the number of search queries by occupation, canton, and day (905 days x 26 cantons x 193 occupations) and is used in a logarithmic transformation for easier interpretation. The treatment variable to estimate the medium-term effect is constructed as a binary variable indicating the time before (=0) and after (=1) the public launch of ChatGPT version 3.5 on the 30th of November 2022.

Other covariates are information on bank holidays and school vacations in each canton. The number of open vacancies is added for each day, canton, and occupation. We also classify occupations according to different dimensions of work content and skills used. First, we classify occupations along their cognitive vs. manual and routine vs. non-routine task intensities (Mihaylov and Tijdens, 2019). Second, for each occupation, we use the official information from schools and career counselors on the cognitive job requirements according to four dimensions: the school language of the region, foreign languages, math, and science.[6] Third, we include the measures on automation risk and exposure to software, robots, and AI in a particular occupation from Frey and Osborne (2017) and Webb (2019), and on exposure to AI and language models from Felten et al. (2023). All available variables can be found in the descriptive statistics in Appendix A.1.[7]

4. Empirical strategy

When OpenAI released ChatGPT for public use, it was a surprise to most people. This is evidenced by the fact that while AI as a generic topic had already been dealt with in the media earlier, media coverage of ChatGPT and what this tool can achieve only started on the day of its launch. The media coverage afterward was rapid and very broad, as the data in Fig. 1a shows. Importantly, media articles often provided practical examples of how well ChatGPT performs various tasks, such as designing an advertising campaign, rewriting a speech, or creating a business plan. The same pattern as for media coverage can be observed for online attention using data from the Google search engine in Fig. 1b. Importantly, no attention was paid to ChatGPT before its launch, providing some more evidence that the launch can be seen as an exogenous shock.

Crucial to our analysis are three observations. First, before the launch of ChatGPT, only very few insiders knew about the possibilities of this and similar AI tools. Second, after the launch of ChatGPT, information about it spread rapidly, and young adolescents learned about its potentially disruptive impact of substituting or complementing work tasks. Third, ChatGPT has no real-time data and thus cannot be used instead of searching for vacancies on an apprenticeship platform. The launch of ChatGPT on November 30, 2022, can, therefore, be seen as an exogenous information shock with a clear before and after. Consequently, the variation we exploit in this study comes from the natural experiment of the unanticipated public launch of ChatGPT.

In the counterfactual world, i.e., in the absence of the launch of ChatGPT, we would expect no difference in search patterns in the post- and pre-treatment phases.[8] To investigate the causal effect in this event study, we must ensure that no other influences related to the timing of the public launch and search patterns in the labor market contaminate our estimated effects. We follow different empirical approaches to analyze the short- and medium-term effects. To analyze the immediate

[3] This proxy for labor supply was first introduced and used by Goller and Wolter (2021). The national online vacancies platform is berufsberatung.ch (German; French: orientation.ch; Italian: orientamento.ch). This is the official platform of the Swiss Conference of Cantonal Ministers of Education for information about and search for apprenticeship positions. There are also other, private platforms, but the national platform collects data on vacancies from employers in all cantons, and is the platform that the majority of apprenticeship seekers rely on during their search and application process.

[4] For the details on the search process on the platform, and why this data is a good proxy for labor supply see Appendix A.2.

[5] Goller and Wolter (2025) show that changes in search queries are linked to changes in the equilibrium of the apprenticeship market. They show that increasing search queries caused by an exogenous reputation shock to some occupations translated into significantly more signed apprenticeship contracts in the year following the shock.

[6] Received from Anforderungsprofile.ch. Because there are no profiles of job requirements available for a few occupations that can be learned through an apprenticeship in Switzerland, the subsample analyses involving this information are on a reduced sample.

[7] Additional information on the data and data preparation can be found in Appendix A.2 and A.3, a list of example occupations according to the used occupational measures in A.4, and a graphical visualization of the search query data in 2021 and 2022 in A.5.

[8] Searches on the vacancies platform are searches for vacancies and not information about occupations, and therefore not influenced by the introduction of ChatGPT itself. The generative AI (at this time) was trained with historical data and had no access to real-time information such as job postings or apprenticeship vacancies. Therefore, even if young adolescents try to substitute the search process with queries on ChatGPT, they still have to search for positions on a vacancies platform.

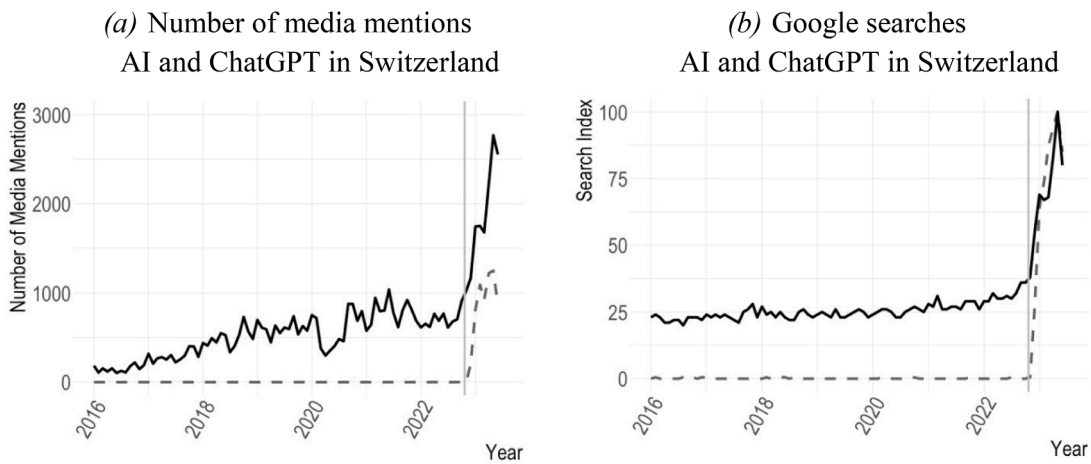

Fig. 1. (a) Number of media mentions AI and ChatGPT in Switzerland, (b) Google searches AI and ChatGPT in Switzerland
Notes: The figure shows the number of media items (Fig. 1a) and the search intensity on the Google search engine (Fig. 1b), from January 2016 through June 2023, by month that includes the terms "AI" (black solid line, in German: "Künstliche Intelligenz") and "ChatGPT" (grey broken line) in Switzerland before and after the introduction of ChatGPT version 3.5 on November 30th, 2022. In Fig. 1b, a Search Index of 100 shows the highest search intensity; lower values represent proportionally smaller numbers of searches. "Media mentions" mainly refer to newspaper articles and radio or television programs for which the transcript is archived in the Swiss media archive. Only the German-language region in Switzerland is considered. The grey line marks the introduction of ChatGPT.

response to the launch of ChatGPT, we use a regression discontinuity (RD) design using the sharp cutoff day (November 30th, 2022). The launch of ChatGPT was exogenous, but not randomized. However, close to the cutoff date, we can exploit the exogenous nature of the launch of ChatGPT as a quasi-experiment; that is, we can treat it as "as good as randomized". The crucial identifying assumption underlying this approach is that nothing else happened around the cutoff date that also influences the outcome. We are aware of only one event around the cutoff date that could bias our results: the Christmas days and the surrounding school vacations, both influencing vacancy search behavior (Goller and Wolter, 2021). To account for this, we restrict the bandwidth around the cutoff date in our preferred specification to three weeks, thus excluding Christmas.

To exclude any shadow effects of Christmas or any other seasonality-related unobserved differential search behavior during this time window, we additionally use the difference-in-discontinuity approach (compare, e.g., Lemieux and Milligan, 2008; Grembi, Nannicini, and Troiano, 2016; Leventer and Nevo, 2024). The general idea is to take the previous year(s) as an unaffected counterpart to construct the difference-in-discontinuity estimator: $\widehat{\tau}_{DD} = (Y^- - Y^+) - (\widetilde{Y}^- - \widetilde{Y}^+)$. The first part is the usual RD estimator that subtracts the boundary estimates before (Y^-) and after (Y^+) the cutoff date. $\widetilde{Y}^-$ and $\widetilde{Y}^+$ are similarly estimated for the unaffected year(s). As suggested in the literature (Gelman and Imbens, 2019), we apply local linear regression in both the usual RD and the difference-in-discontinuity. With this, the latter can be formalized as follows:

$$Y_{itcp} = \delta_0 + \delta_1 D_{it} + P_i(\gamma_0 + \gamma_1 D_{it}) + T_t[\alpha_0 + \alpha_1 D_{it} + P_i(\beta_0 + \beta_1 D_{it})] + \varepsilon_{itcp},$$

where Y_{itcp} contains the (log) number of search queries by canton (c), occupation (p), day (i), and year (t). T_t is a binary indicator with a value of 1 for the year 2022 and 0 otherwise. P_i equals 1 for the days after the cutoff date (D^{cut}) and 0 before. $D_{it} \in [D^{cut} - h, D^{cut} + h]$ contains the days before and after November 30 (D^{cut}), with bandwidth $h = 14, 21, 28$ days; ε_{itcp} is an idiosyncratic error term. β_0 is the difference-in-discontinuity estimator. The implicit assumption of this approach is that the search behavior for the affected and unaffected year(s) follows the same pattern, except for the effect following the launch of ChatGPT. We therefore excluded the year 2020 from this analysis, because the COVID-19 pandemic and the politically enforced measures disturbed the vacancy search behavior (Goller and Wolter, 2021).

For their assumptions to be credible, discontinuity-based strategies estimate effects close to the cutoff. Therefore, we use a different empirical strategy to estimate medium-term effects, relying on a conditional independence assumption. Most importantly, during a year, searches for apprenticeship vacancies follow some cyclical pattern resulting from the apprenticeship market's clearing mechanisms. Thus, we use a month fixed-effect model to ensure that we estimate the effect for the comparable time in the search cycle. The approach can be formalized as follows:

$$Y_{cpd} = \alpha + \beta W_d + \gamma X_{cpd} + \delta_{(p)m} + \eta_t + \zeta_c + \varepsilon_{cpd}$$

W_d is the treatment variable that equals 1 after day d (November 30, 2022), the day of the launch of ChatGPT, and 0 before. Besides the month fixed-effects (δ_m) or month-by-occupation fixed effects (δ_{pm}), we take several potential confounders into account. The maximum supply of apprentices is determined by yearly changing cohort sizes that potentially enter the apprenticeship market; consequently, we control for the year (η_t). Because of potentially different perceptions about the capability of generative AI in the different Swiss regions, and different search behavior, we control for the cantons (ζ_c). Moreover, captured in X_{cpd}, we control for days of the week, school vacations, and public holidays. This might be particularly important in this study because of the Christmas holidays and accompanying school vacations shortly after the announcement of ChatGPT, which otherwise could lead to confounded effects if they differ from the previous years' vacations. Standard errors are, in all cases, clustered on the occupational level.

We assess the credibility of this specification to estimate causal effects in multiple ways. First, we assess the sensitivity toward including different covariates in the model. Particularly, to ensure that potentially different cyclical search patterns for different occupations do not bias our results, we check the sensitivity towards including month-by-occupation fixed effects. Second, we use different samples and time periods to estimate the effects in robustness checks. Third, we investigate whether the demand side influences the effect, i.e., by offering more or fewer vacancies. Fourth, in a placebo treatment exercise, we study a

pseudo-treatment for the days/weeks before the public deployment of ChatGPT.

5. Empirical results

We investigate how the unexpected launch of ChatGPT in November 2022 affected young people's search behavior for apprenticeship vacancies. We present results for the immediate, short-term effects in Section 5.1, followed by the medium-term and heterogeneous effects in Section 5.2. Section 5.3 investigates how the short- and medium-term effects on supply affect the apprenticeship market in the equilibrium. Section 5.3.1 shows the influence of the launch of ChatGPT on the number of signed apprenticeship contracts, and Section 5.3.2 presents a case study on the commercial employee, the by far most widely offered apprenticeship occupation, and at the same time one of the most affected occupations by ChatGPT.

5.1. Short-Term effects

Fig. 2a presents the results of the regression discontinuity estimation for the sharp discontinuity on November 30, 2022. We see a drop in search queries after the discontinuity, with a point estimate of −0.098, which is statistically significant. In contrast, when investigating the same time frame in 2021, we find no differential effect. Fig. 2b shows no statistical difference and an estimate close to zero (0.009).[9] We can conclude from these analyses that the launch of ChatGPT negatively influenced our proxy of labor supply in the period just after its launch.

Two primary reasons can cause such a decline. First, an inherent decline in search queries due to a change in preferences of the labor supply. Second, a decline in firms' demand for apprentices. In Fig. 2c, we investigate the number of posted vacancies around the launch of ChatGPT. We find no difference on the demand side, i.e., the number of vacancies in 2022 is unaffected by the launch of ChatGPT. This aligns with our expectations because a) firms are usually long-term oriented and do not change their plan short-handed, and b) firms have several ways to adjust to a technological change, such as changing the curricula for the apprentices to educate them to their needs or hiring additional specialists on the regular labor market when needed. The decision to continue offering the apprenticeship positions is, therefore, a low-stakes decision, while it is a high-stakes decision for the individuals choosing to learn specific skills.

Table 1, Panel A, shows the regression discontinuity estimates for different bandwidth choices. The estimates are relatively insensitive towards choosing 2, 3, or 4 weeks around November 30. To complete the analysis of the short-term effects, Table 1, Panel B shows the difference-in-discontinuity estimates. In the difference-in-discontinuity, we take the unaffected time frame around November 30, 2021, as a counterfactual. For the various bandwidth choices, compared to the RD estimates, the effect sizes are slightly lower in magnitude; however, we see a decline in labor supply by about 0.08. As this is a log transformation, we can interpret this as a decrease of 8 percentage points, an economically highly relevant effect that is statistically significant at all conventional significance levels.

5.2. Medium-Term effects

Table 2 shows the medium-term impact of the shock on perceptions about the capabilities of AI substituting human tasks, measured by the introduction of ChatGPT on the supply of apprentices, with and without the inclusion of potential confounders. Controlling for the most important confounders, i.e., month, year, and occupation fixed effects (column 2), we find a coefficient of −0.082, which can be interpreted as 8.2 % fewer search queries following the public launch of ChatGPT.[10] We find the medium-term impact to be very similar to the short-term effect that we estimated in Section 5.1.

The result is robust to including additional controls in columns 3 and 4.[11] Further robustness checks (see Appendix B.2) show that the effect is also robust to different sample specifications and to controlling for the number of vacancies (the demand side). Moreover, in contrast to the supply of potential apprentices, we do not find a statistically significant impact of the introduction of ChatGPT on the number of vacancies in the medium term. Additionally, in Appendix B.3, we show the medium-term temporal variation of the impact in the weeks that immediately followed the introduction of ChatGPT. Importantly, for the weeks before the launch, conducting a placebo-treatment estimation, we find a zero effect of the pseudo-treatment, i.e., no anticipation effect. After the launch, the effect is immediately visible, and the magnitude of the effect increases over the first weeks. While the effect sizes decrease after their peak in weeks 5/6, the medium-term effects shown in Table 2 indicate that the effect lasts over the whole observation period (until June 2023). This is in line with Fig. 1, showing the number of media mentions of ChatGPT that increased in the months after its launch and remained at a high level until the end of June 2023.

5.2.1. Task type intensity

The impact on the aggregated searches for apprenticeship vacancies shows an effect that might be understood as a general shock because young people looking for an apprenticeship do not know what educational or occupational path to choose. On the other hand, it is evident that the possibilities opened up by generative AI, such as ChatGPT, with regard to the substitution (or complementation) of work tasks affect individual occupations very differently (Hui et al., 2024; Demirci et al., 2025). For this reason, we turn to the potential heterogeneity of the effect according to different types of occupations. To investigate this, we conduct our analyses separately for occupations that predominantly contain certain task types that are more or less in danger of skill obsolescence due to generative AI.

Table 3 shows the estimates for different subsamples of occupation types classified as predominantly intensive in routine versus non-routine and manual versus cognitive tasks. While the effects on the search for predominantly routine vs. non-routine occupations are almost identical (WALD test for equality, p-value: 0.68), we find a significantly stronger negative effect on the search for more cognitively demanding occupations compared to manually intensive ones (WALD test for equality, p-value: 0.00). Combining the two dimensions to classify occupations into four types, we find by far the largest decline in searches for routine cognitive occupations.

While this subchapter focuses on the medium-term effects, short-term RD and difference-in-discontinuity, estimates by occupation type can be found in Appendix B.4. Similar to the average effects, short and medium-term effects are almost equal for occupations with primarily manual or cognitive tasks.

[9] We conducted a placebo regression discontinuity in the year 2020, which can be found in Appendix B.4. Also, in this estimation there is no differential effect.

[10] In a recent article, Chen and Roth (2024) found that for log-like outcome transformations, estimated coefficients may not be invariant to the unit of the outcome, while a percentage interpretation should be unit-independent. For our study, this implies that the percentage interpretation of the estimated effects depends on the baseline units in which our outcome is measured. As an alternative to the log-like transformation, in Appendix B.1, we used one of the proposals of Chen and Roth (2024) and replicated Table 2 using a Pseudo Poisson Maximum Likelihood (PPML) estimation (Santos Silva & Tenreyro, 2006). The results, although in levels and not in percentages, are in line with the main results.

[11] We use model 3 as our baseline specification in the following analyses.

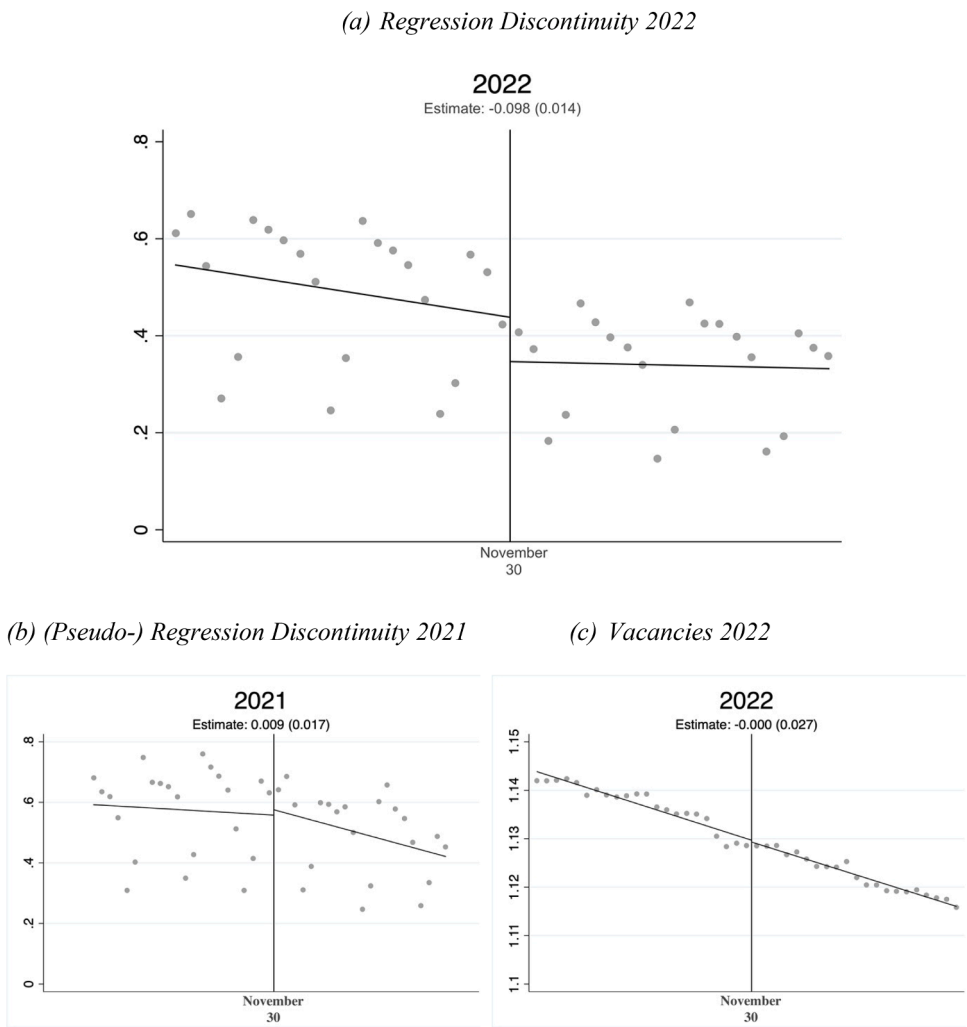

Fig. 2. *(a) Regression Discontinuity 2022, (b) (Pseudo-) Regression Discontinuity 2021, (c) Vacancies 2022*
Notes: RD estimations. The bandwidth is 21 days (3 weeks). The vertical line marks the (pseudo-) introduction of ChatGPT on November 30, 2022 (2021). Points in subfigure (a) and (b) represent the log search queries averaged by day, occupation, and canton. Points in subfigure (c) represent the log vacancies averaged by day, occupation, and canton. The two lines before and after November 30, represent two separate local linear regressions. The standard errors for the RD estimates are clustered on the occupational level (in parentheses).

Table 1
Short-term effect of the launch of ChatGPT on labor supply.

	(1)	(2)	(3)
Panel A: Regression Discontinuity			
ChatGPT	−0.105	−0.098	−0.098
	(0.014)	(0.014)	(0.014)
Bandwidth	2	3	4
N	141,232	211,848	282,464
Panel B: Difference-in-Discontinuity			
ChatGPT	−0.098	−0.083	−0.081
	(0.020)	(0.020)	(0.020)
Bandwidth	2	3	4
N	282,464	423,696	564,928

Notes: The outcome is the log of the number of search queries by day, canton, and occupation. Panel B includes controls for the day of the week. Standard errors are clustered on the occupation level.

Fig. 3 shows how the effects for manual and cognitive occupations evolve over the first 14 weeks after the public launch of ChatGPT (vertical line). Although both categories of occupations experience a decrease in the search activity for vacancies, there is a statistically

Table 2
Medium-term effects of the launch of ChatGPT on labor supply; sensitivity of effects.

	(1)	(2)	(3)	(4)
ChatGPT	−0.063 (0.009)	−0.082	−0.087	−0.086
		(0.007)	(0.007)	(0.007)
School vacation			−0.110	−0.096
			(0.008)	(0.007)
Public holiday			−0.124	−0.115
			(0.010)	(0.009)
Month FE		X	X	
Year FE		X	X	X
Occupation FE		X	X	
Day of the week			X	X
Canton FE				X
Occupation by month FE				X
Sample	01.01.2021 – 30.06.2023			
N	4,541,290	4,541,290	4,541,290	4,541,290

Notes: Linear regression. The outcome is the log of the number of search queries by day, canton, and occupation. Standard errors are clustered on the occupation level. Every regression includes a binary indicator for the days the platform was the victim of a hacker attack.

Table 3
Subsamples by occupation types.

	(1) Routine	(2) Non-Routine	(3) Manual	(4) Cognitive
ChatGPT	−0.091 (0.016)	−0.084 (0.007)	−0.062 (0.005)	−0.147 (0.019)
N	1,623,570	2,917,720	3,200,080	1,341,210
	Routine Manual	Routine Cognitive	Non-Routine Manual	Non-Routine Cognitive
ChatGPT	−0.051 (0.007)	−0.198 (0.045)	−0.068 (0.006)	−0.121 (0.018)
N	1,176,500	470,600	2,023,580	894,140

Notes: Linear regression. The outcome is the log of the number of search queries by day, canton, and occupation. Standard errors are clustered on the occupation level. Each regression uses the same specification as in Table 1, column (3).

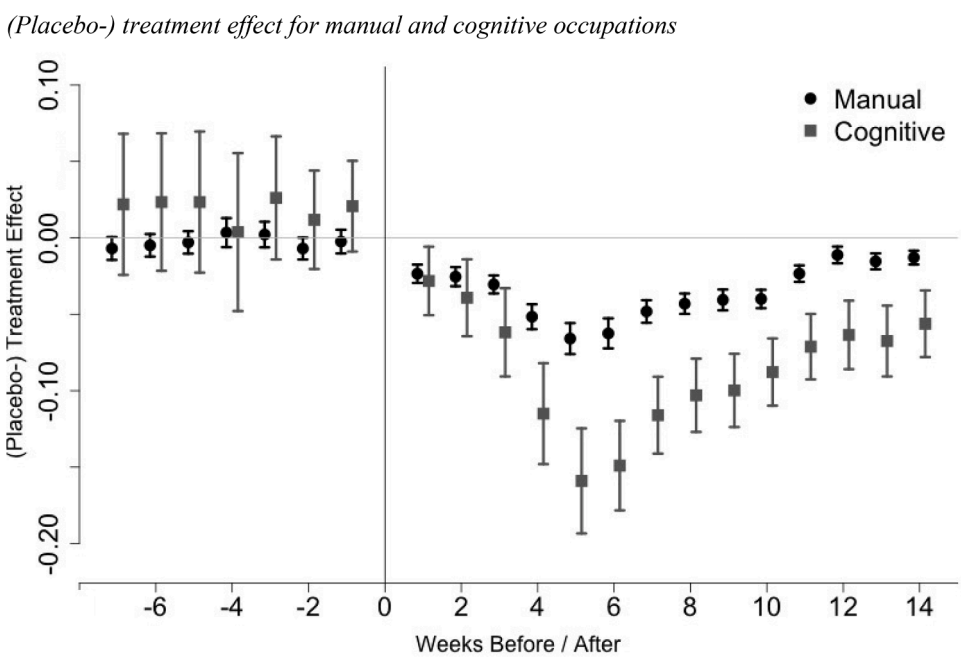

Fig. 3. (Placebo-) treatment effect for manual and cognitive occupations
Notes: The figure shows the (placebo-)treatment effect for manual versus cognitive occupations before and after the introduction of ChatGPT (in $t = 0$). Occupations are categorized depending on whether they primarily involve manual or cognitive tasks (based on Mihaylov and Tijdens, 2019). On the left side, placebo treatment effects for the x weeks before the introduction are estimated. Each point refers to a point estimate of a separate linear regression (using column (3) model specification, Table 2). The whiskers mark the 95 % confidence intervals, with standard errors clustered on the occupation level.

significant difference in the effect size after the 4th week. The overall negative effect can be seen over the entire time span of 14 weeks following the launch of ChatGPT. Before the introduction of ChatGPT, both estimates were close to zero. This indicates that there was no anticipation effect before the event, and both types of occupations followed similar pre-trends.

5.2.2. *Technology exposure and occupational requirements*

To complete the picture of which occupations are most affected, we classify occupations according to their cognitive requirements, as well as current and previous measures for technology exposures. We differentiate the impact of ChatGPT for low, medium, and high requirements in the school language, the risk of substitution by automation as proposed by Frey and Osborne (2017), exposure to robots by Webb (2019), and the latest exposure measures to large language models by Felten et al. (2023).

Table 4 provides the results for the estimations for occupations in the respective requirements, risk, or exposure subsets.[12] For the breakdown of occupations according to the risk of substitution by automation (Frey and Osborne, 2017), we find the largest declines in the supply of apprentices for those occupations with the lowest assigned automation risk. A similar pattern is found for the occupations classified by their exposure to robots (Webb, 2019).

However, using the latest measures from Felten et al. (2023) of language model exposure aligns with the revealed preferences of the search activities. Occupations with high exposure to language models show by far the largest decline in interest, and there is a lower adverse effect for both medium and low exposure categories. In line with this, and with ex-ante expectations, because ChatGPT is, at first sight, useful

[12] In Appendix B.5, we additionally report subsample results for estimates on exposure to software and AI (Webb, 2019) and AI (Felten et al., 2023), and requirements in foreign languages, math, science, and a combined measure of competencies (overall requirements). Notably, we used the crosswalk provided by Hardy et al. (2018) to assign these measures to the apprenticeship occupations. A list of exemplary occupations in the classifications and categories of Table 4, can be found in Appendix A.4, Table 8.

Table 4
Subsample effects, technology exposure and occupational requirements.

	(1) low	(2) medium	(3) high
Automation risk (Frey and Osborne, 2017)			
ChatGPT	−0.151	−0.084	−0.073
	(0.020)	(0.010)	(0.010)
Robot exposure measure (Webb, 2019)			
ChatGPT	−0.132	−0.078	−0.062
	(0.021)	(0.010)	(0.008)
LM exposure measure (Felten et al., 2023)			
ChatGPT	−0.054	−0.068	−0.140
	(0.006)	(0.010)	(0.017)
Occupational requirements in school language skills[a)]			
ChatGPT	−0.068	−0.068	−0.142
	(0.010)	(0.008)	(0.022)

Notes: Linear regressions on the outcome, the log of the number of search queries by day, canton, and occupation. In the three columns, subsamples are analyzed by splitting the sample into low, medium, and high, each about 33 % of observations according to the requirements/exposure. Standard errors are clustered on the occupation level.

[a] Here, a reduced sample of occupations is used for which we observe the occupational requirements ($N = 3576{,}560$). Requirements are weighted by their relevance for each occupation.

Table 5
Difference-in-differences estimation of the launch of ChatGPT on the number of signed apprenticeship contracts.

	(1)	(2)	(3)	(4)	(5)
Panel A: Difference-in-Difference – Cognitive vs. *manual*					
Cognitive x Year 2023	−0.054	−0.085	−0.004	−0.044	−0.027
	(0.155)	(0.127)	(0.127)	(0.144)	(0.100)
Cognitive	0.203	0.497	−0.009	0.207	0.016
	(0.162)	(0.127)	(0.137)	(0.153)	(0.028)
Year 2023	0.033	0.056	0.016	0.017	0.019
	(0.035)	(0.024)	(0.029)	(0.032)	(0.018)
Panel B: Difference-in-Difference – Highly exposed to LM vs. *medium/low exposure*					
High LM exposure x Year 2023	−0.003	−0.043	0.030	−0.001	0.003
	(0.143)	(0.117)	(0.117)	(0.133)	(0.092)
High LM exposure	0.181	0.586	−0.124	0.185	0.152
	(0.160)	(0.122)	(0.135)	(0.151)	(0.097)
Year 2023	0.012	0.041	0.001	−0.000	0.039
	(0.029)	(0.014)	(0.027)	(0.026)	(0.019)
Sample	all	female	male	Swiss	Non-Swiss
N	54,626	54,626	54,626	54,626	54,626

Notes: The outcome is the log of the number of signed apprenticeship contracts (column 1). Columns (2)-(5) use the log of the number of signed apprenticeship contracts for the subpopulation that is indicated in the row "Sample", i.e., female, male, Swiss, non-Swiss. Standard errors are clustered at the occupation level.

for language-related tasks, the decrease in search queries for jobs with the highest language requirements is the largest.[13]

5.3. Equilibrium effects

Sections 5.1 and 5.2 establish a general decrease in the supply of apprentices, particularly for occupations with primarily cognitive tasks. We also know that demand remained unaffected by the launch of ChatGPT. For the equilibrium, this leaves two options: a decreasing quantity or quality of apprentices, i.e., either the firms do not fill all their vacancies, or they fill most of their vacancies but take candidates with lower ability. In Subsections 5.3.1 and 5.3.2, we present descriptive analyses on potential equilibrium effects using supplementary data sets that we describe in more detail in Appendix C.

5.3.1. Number of Apprenticeship Contracts

In order to assess the quantitative impact on the apprenticeship market, we use administrative data on apprenticeship contracts. Table 5 shows two difference-in-differences estimations, using occupations with primarily cognitive tasks (Panel A) or occupations with high LM exposure (Panel B; measure based on Felten et al. (2023)) as treated occupations. Observing the time frame from 2013 to 2023, the period that was 'affected' by the launch of ChatGPT was the year 2023. The interaction of the indicator for the treated occupations ("Cognitive" or "High LM exposure") with the binary indicator for the observations that are affected in time ("Year 2023") is the difference-in-difference estimator.

In column (1), Table 5, we investigate the overall effect. If anything, the effect is negative, but neither in Panel A nor in Panel B are the estimated effects statistically different from zero. In columns (2)-(5), we additionally find that the numbers of apprenticeship contracts for female (2), male (3), Swiss (4), and non-Swiss (5) are not affected. In total, we find a negative, but statistically insignificant, effect of the launch of ChatGPT on the number of apprenticeship contracts for occupations that are potentially highly exposed to generative AI, and for which we found a decrease in supply.

5.3.2. Case study: commercial employee

If there is no statistically significant effect of the decrease in the supply of apprentices on the quantitative equilibrium level, i.e., the number of signed apprenticeship contracts, the question arises whether the launch of ChatGPT did affect the apprenticeship market at all. In this subsection, we address this question by looking at changes in the quality of applicants in response to the launch of ChatGPT. To investigate this question, we collected additional data from the largest company that offers standardized occupational aptitude tests for people who plan to apply for apprenticeship vacancies. It is common practice in Switzerland to add the certificate that shows the test result of this standardized test to the application.

In this case study, we compare the test results from the occupational aptitude test of those who planned to apply for the apprenticeship in the occupation that was among the most affected in terms of a decrease in supply: the commercial employee. The occupation of commercial employee is particularly suitable for a case study because it is the largest and most popular apprenticeship in Switzerland. For counterfactual comparison, we did the same for those who wanted to apply for any other occupation.[14]

Interestingly, Table 6 shows that the skills measured in different subjects in school knowledge and occupational knowledge decreased for those who planned to apply to the occupation of a commercial employee, comparing those who did the test before the launch of ChatGPT to those who did the test after the launch. The counterfactual analysis shows that none of the test scores changed for those who had taken the test for any other occupation before and after the launch of ChatGPT.

We conclude that the quality of applicants decreases in particularly exposed occupations, i.e., if we look at the skills of people interested in starting an apprenticeship in an occupation that is perceived to be highly affected by generative AI. This might be because the more capable applicants, aware of the potential risk of having to learn skills in a

[13] Cognitive requirements are measured from 1 to 100, where 1 is the least demanding, and 100 is the most demanding. The classification of apprenticeships was carried out by experts in the Conference of Cantonal Ministers of Education and the Swiss Trade Association.

[14] The analysis relies on the exogenous nature of the launch of ChatGPT, and control for gender, test language, test year, month, and occupation. However, we regard the analysis as descriptive evidence, not claiming causality.

Table 6
Event study on the launch of ChatGPT test scores in an occupational aptitude test.

	Test-Scores occupational aptitude test	
	Commercial Employee	All (without Commercial Employee)
School knowledge	−0.898 (0.272)	0.699 (0.745)
First language	−0.407 (0.505)	1.135 (0.950)
English language	−1.744 (0.392)	0.974 (0.607)
Math	−0.227 (0.403)	0.186 (1.087)
Occupational knowledge	−1.672 (0.568)	0.955 (0.757)
Organization	−1.875 (0.867)	0.026 (0.953)
Connected thinking	−1.566 (0.655)	1.429 (1.571)

Note: Test scores before and after the launch of ChatGPT in multiple categories. The first column contains estimates for tests conducted by people who have planned to apply to the occupation of a commercial employee; the second column contains estimates for those who have planned to apply to any other occupation. Each estimation contains control variables: gender, test language, test year, month, and an indicator for the occupation (column 2 only). Robust standard errors are in parentheses.

particular occupation that will become obsolete due to AI, try to move to other occupations that have a lower degree of exposure to AI.

6. Discussion and conclusions

In this paper, we present the first evidence of the influence of the availability of generative AI on the career choice behavior of young people. In the first place, we show that the launch of ChatGPT led to a short- and medium-term slump in search queries for apprenticeship vacancies in general. This could reflect an increased drive for more general education or an uncertainty about which occupations teach more future-proof skills. While both explanations might be part of the solution, we can show that the individual occupations were affected by the slump to very different degrees. As expected, the decline affected occupations with a high proportion of cognitive tasks more than occupations with a high proportion of manual tasks. Similarly, occupations with high language skill requirements experienced a significantly greater impact than those with high demands in mathematical skills. Compared to earlier estimates on the substitutability and exposure of jobs by computer, robot, or AI technology (Frey and Osborne, 2017; and Webb, 2019), the results ultimately indicate that search queries fell less sharply in occupations traditionally considered highly exposed.

In contrast, the impact on search activity is entirely consistent with more recent estimates of the impact of AI on occupations (Felten et al., 2023), i.e., occupations that are highly exposed to AI technology in general and language models in particular were also significantly more affected by the decline in search queries. This indicates that the way young people see AI, as far as the impact on specific occupations is concerned, is in line with the most recent expert opinions; however, independently of the experts, as the young people were not aware of these expert opinions at the time of the introduction of ChatGPT. In other words, young people also assume that generative AI will now affect occupations that were previously considered low risk, rather than those previously considered high risk, and that generative AI could be seen as a "game changer" in the interplay between digital technology and developments in the labor market – especially since, for the first time, such a transformative technology was immediately and freely accessible for hands-on use by anyone with internet access. Furthermore, Cattaneo et al. (2025) found that individuals have a strong interest in reducing their exposure to automation risks and are willing to accept significantly lower wages. In this sense, our findings might indicate that young people aim to minimize automation risks in their occupational choices from the start.

Only time will tell whether the young people who are faced with a high-stakes decision, that is, in which occupation-specific skills they want to invest in the next three to four years of training, are right or wrong in their assessments. At the very least, we can now see that they have radically changed their assessments of how specific occupations will be affected by the possibilities of AI through the use of tools such as ChatGPT – a shift that aligns with the expectations of experts who anticipate that rapid advances in AI will fundamentally reshape labor markets and require workers to adapt quickly to new realities.

To shed some light on the tangible effects of the decrease in labor supply, we provide two additional descriptive analyses. First, we find a rather negative, although statistically insignificant, correlation between the launch of ChatGPT and the number of apprenticeship contracts for occupations with primarily cognitive tasks. Second, we find a negative correlation between the launch of ChatGPT and the ability of the individuals in the applicant pool for the occupation of a commercial employee, the most popular occupation, which is at the same time likely one of the most affected by generative AI. In this sense, the introduction of ChatGPT may very well have tangible effects on the labor supply of particularly affected occupations, also contingent on long-term changes in occupational choices – a question that remains beyond the scope of this study. While further research will be needed to assess the long-term shifts in the apprenticeship market, it is clear that this time, it's different: the unprecedented, rapid and free availability of generative AI has both shifted automation risks toward cognitive and previously low-risk occupations and rapidly reshaped labor expectations and career choices among young people – distinct from what was seen in previous technological revolutions.

CRediT authorship contribution statement

Daniel Goller: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Christian Gschwendt:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Conceptualization. **Stefan C. Wolter:** Writing – review & editing, Writing – original draft, Supervision, Resources, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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Data availability

The data that has been used is confidential.

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